

$$\frac{r^2}{(D-r)^2} = \frac{1}{16}$$

Q4(a)(i)

When light shone on LDR, resistance of LDR is $420\ \Omega$.

$$\text{p.d across LDR} = \frac{420}{420+8200} \times (4.0 - (-2.0)) = 0.29 \text{ V}$$

$$\text{Hence potential at P, } V_P = 0.29 - 2.0 = \mathbf{-1.71 \text{ V}}$$

4(a)(ii)

When LDR in the dark, resistance of LDR = 134 kΩ.

$$\text{p.d across LDR} = \frac{134 \text{ k}\Omega}{134 \text{ k}\Omega + 8.2 \text{ k}\Omega} \times 6.0 = 5.65 \text{ V}$$

$$\text{Hence potential at P, } V_P = 5.65 - 2.0 = \mathbf{3.65 \text{ V.}}$$

4(a)(iii)

The alarm is placed in parallel with the LDR. When the laser beam is shining on the LDR, the resistance of the LDR will be low. By the potential divider principle, the p.d. across the LDR and alarm will be low. The alarm will remain turned off.

When the burglar blocks the laser beam, the resistance of the LDR will be high. Hence the p.d across the LDR and alarm will be high. This will turn on the alarm.

4(b)(i)

There will be no deflection in the galvanometer when the p.d across JY = 1.5 V.

Let the balance length (between J and Y) be L . Then

$$V_{JY} = \left(\frac{L}{1.00} \right) \frac{1.50}{(1.50+0.50)} \times 4.0 = 1.5$$

$$L = 0.500 \text{ m.}$$

$$\text{Hence distance from X} = 1.00 - L_{JY} = \mathbf{0.500 \text{ m.}}$$

4(b)(ii)

When 1.50 Ω connected with the 1.5 V cell, the terminal p.d across the 1.50 Ω is $V =$

$$\frac{1.50}{1.50+0.50} \times 1.5 = 1.13 \text{ V}$$

There will null deflection when $V_{JY} = 1.13 \text{ V.}$ Then

$$V_{JY} = \left(\frac{L'}{1.00} \right) \frac{1.50}{(1.50+0.50)} \times 4.0 = 1.13$$

$$L' = 0.377 \text{ m.}$$

$$\text{Hence distance from X} = 1.00 - 0.377 = \mathbf{0.623 \text{ m.}}$$